STANDARD OPERATING PROCEDURE OF ROSS FLEX VF-D02

Lab /SOP/ 20



Ross flex machine

Digital calipers or Steel scale

Piercing tool B with cutting edge of 2.5±0.02 mm in length



Die-cut test samples 2.54cm in width by 15.2cm in length.

Solid Rubber 3.0 ± 0.2 mm

TPU, Polyester (Hytrell), Nylon 3.0 ± 0.2 mm

Microcellular Rubber, EVA, PU 5.0 ± 0.2 mm

Thermoplastic rubber and PVC 7.0 ± 0.2 mm

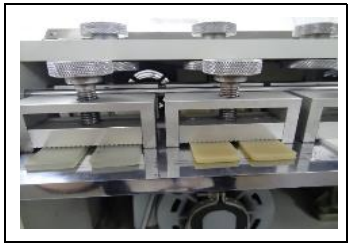
Heel counter as supplied



Position the test sample in the piercing tool such that the chisel tip is located over the center of the test sample and 62mm from one of the ends

Push the chisel tip just far enough into the test sample so that a 2.5mm wide slit is produced throughout the entire thickness of the sample.

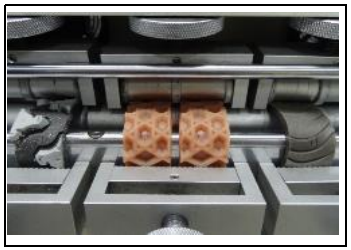
Mark the 2.5mm slit and measure it by cliper.



Raise the holder arm of the Ross Flex machine to the horizontal position.

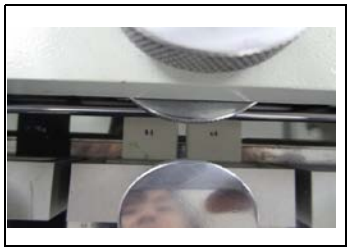
Insert the test sample through the clamp on the holder arm and the top rollers at the rear of the machine.

Clamp the sample in the holder arm.



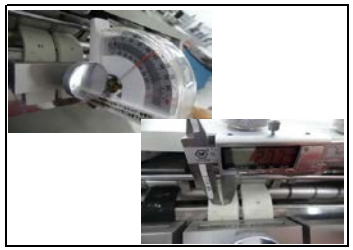
Lower the top rollers until they just touch the test sample, and lock in this position using the wing nut.

Zero the counter and start the machine.



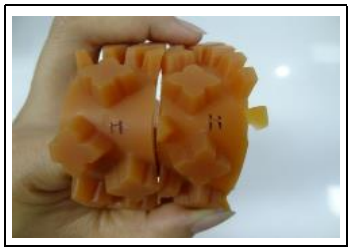
If the sample is to be run for a maximum of 100,000 cycles, check at 10,000, 25,000, 50,000, and 100,000 cycles.

In case a shorter test is to be run, for example at 25,000 cycles, the sample should be checked at every 5,000 cycles.



Failure is defined as “when the slit has grown to a length of 5.0mm”. This evaluation should be made with the folder arm at an angle of approximately 45 degrees.

If the crack growth is not in a straight line, measure the distance of the longest continuous crack, regardless of its direction.



Cut growth (%): (Lf - Li) / Li *100

Where: Li = 2.5 mm Lf = Final cut growth

No slit should be made in cellulose insole boards or painted midsoles.

Prepared by	Approved by	REVISION		DATE	